

Exit Meeting

08/29

Hyunseo Lee

KAIST Visual AI Group

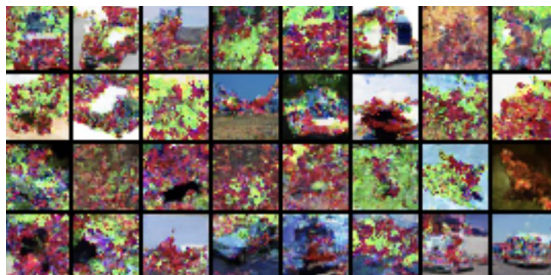
Discrete Flow Matching

Cifar10

30epochs



200epochs



Guided results

Mnist

1epochs



30epochs

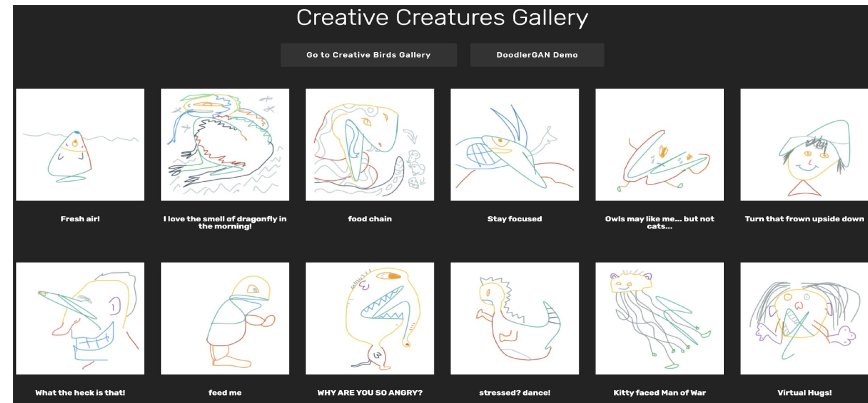


What I'm doing

Quickdraw Dataset



Creative Sketch Dataset



More complicated task
Has either class label and stroke label

Input: sequence of points (sketches)

output: reconstructed sketch

Goal is to Construct the Sketch Generation Model

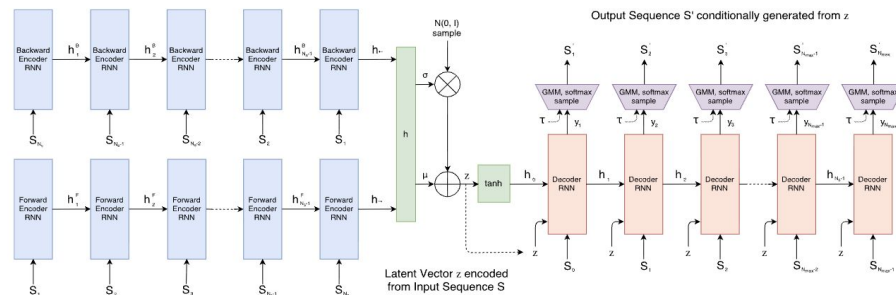
Previous work

ICLR 2018

A Neural Representation of Sketch Drawings

David Ha
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Douglas Eck
Google Brain
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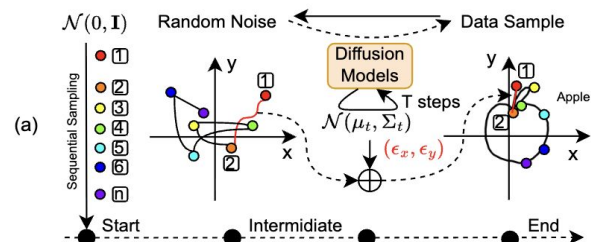


ICLR 2023 (notable top-25%)

SKETCHKNITTER: VECTORIZED SKETCH GENERATION WITH DIFFUSION MODELS

Qiang Wang¹ Haoge Deng¹ Yonggang Qi^{1*} Da Li² Yi-Zhe Song²

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Previous work

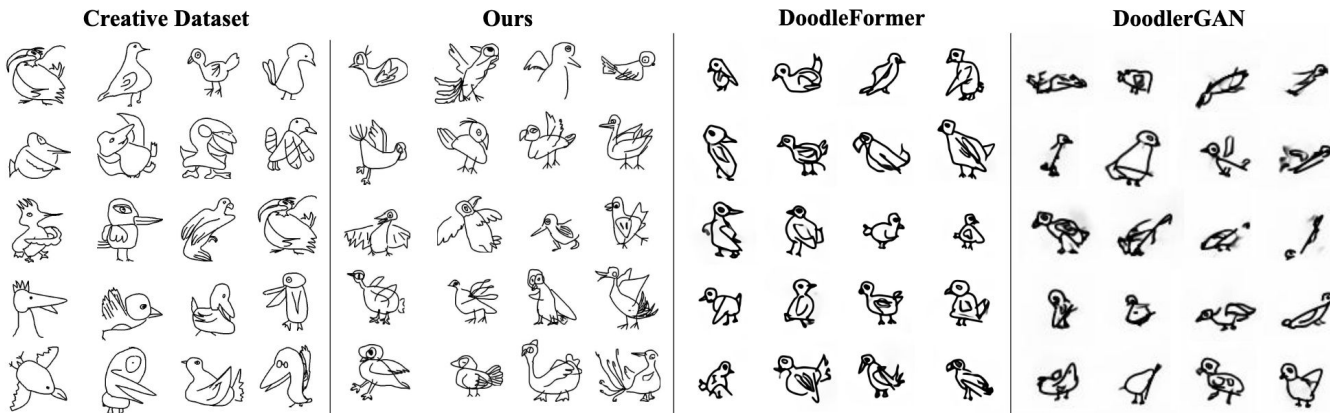
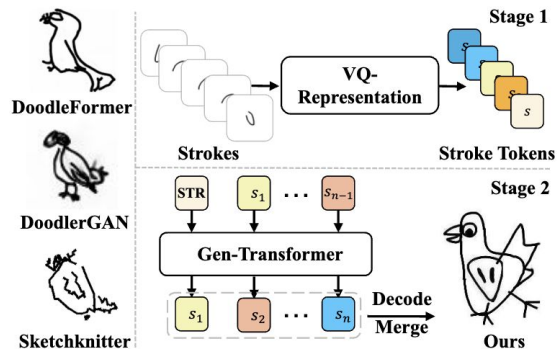
ICCV 2025

VQ-SGen: A Vector Quantized Stroke Representation for Creative Sketch Generation

Jiawei Wang^{1,2} Zhiming Cui² Changjian Li¹

¹The University of Edinburgh ²ShanghaiTech University

VQ(Representation) + Transformer(Generation)



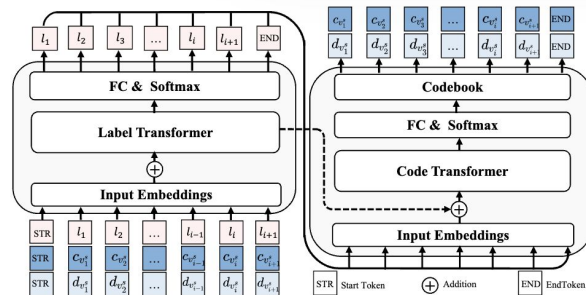
Previous work

Key limitation

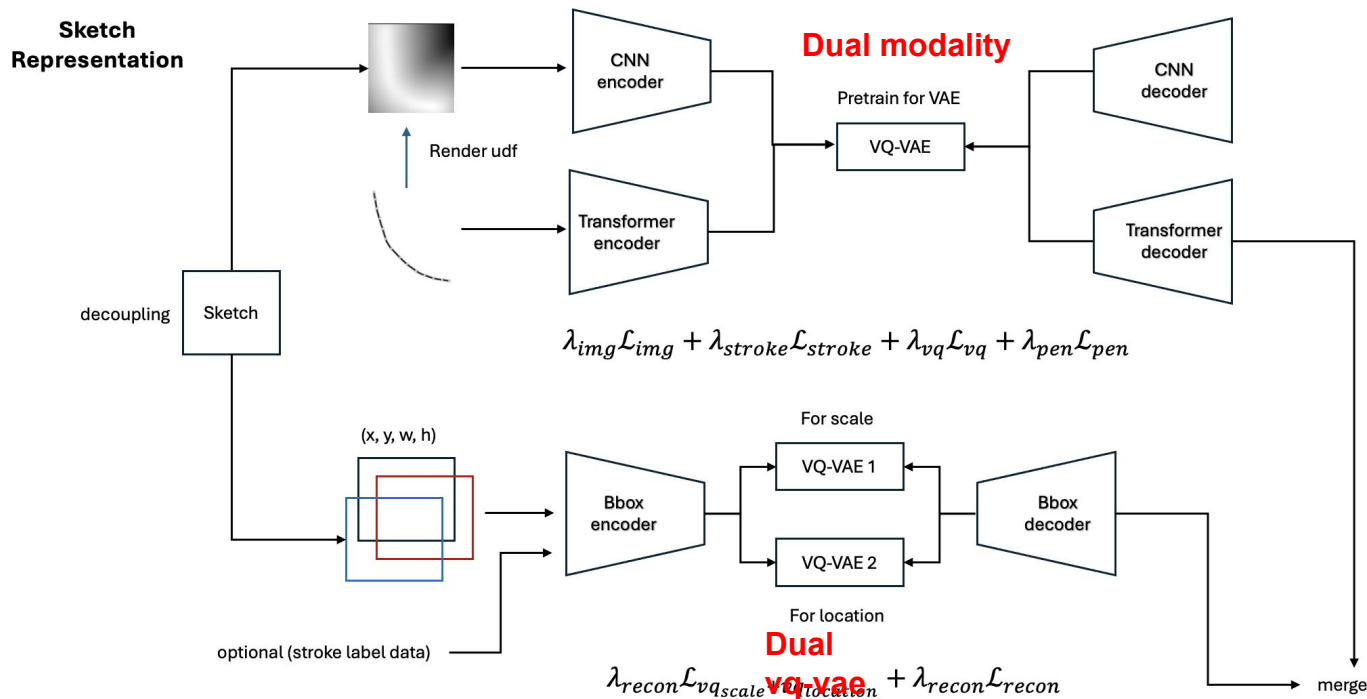
- (-) Transformer can't change its decision and have to generate sequentially!
- (-) It's task is very limited

Objective

- (+) Build more General model for a many sketch task

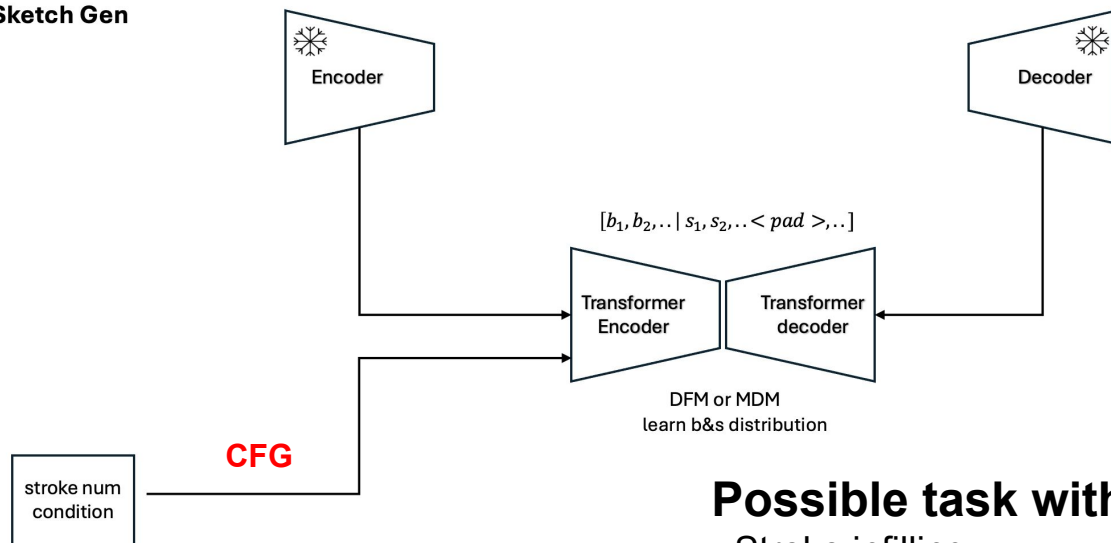


My approach



My approach

Sketch Gen



Possible task with (DFM or MDM)

- Stroke infilling
- Move stroke to arbitrary position and repaint
- Layout generation with given stroke or stroke generation with given layout

Expected advantage

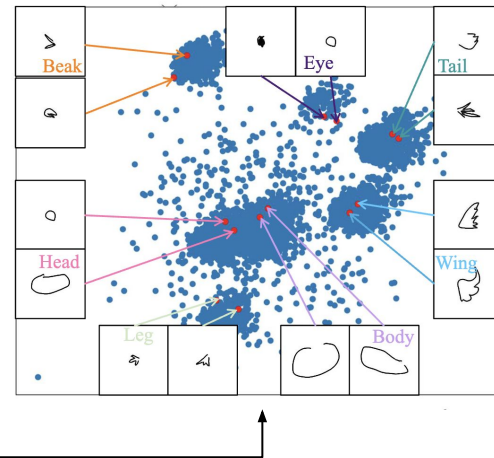
Why VQ?

- (+) it can substitute sketch generation problem to text generation task
- (+) it can ignore some unnecessary stroke and learn more superficial, general representation
- (+) with VQ-VAE it can generate good latent space

MDM(Diffusion) vs DFM(Flow)

- (+ MDM) it is good for logical inference like sudoku
- (+ DFM) it is good for general task like code infilling and high NFE task
- (+ Both) it can leverage global context and generate parallel

Maybe MDM is good for my case..
I will ablation for both MDM, DFM!



There are constraint condition
eyes, mouse, eyebrow



Sketch is like logical
puzzle!

Experimental setup

Sketch Representation

```
config = {
  "max_stroke_len": 48,
  "udf_resolution": 64,
  "d_model": 256,
  "d_img": 128,
  "d_seq": 128,
  "embedding_dim": 256,
  "num_embeddings": 293,
  "batch_size": 256,
  "learning_rate": 1e-4,
  "1-epochs": 1000,
  "2-epochs": 1500,
  "3-epochs": 1000,
  "gamma": 100.0,
  "rdp_epsilon": 0.0,
  "commitment_cost": 0.25,
  "n_head": 8,
  "num_layers": 6,
  "overfit_size": 10,
  "data_size_per_category": 50,
  "loss_weight": {
    "coord": 1.0,
    "pen": 0.1,
    "img": 0.1,
    "vq": 0.8,
    "kl": 0.01
  },
  "image_num_layer": 4,
  "bbox_hidden_dim": 64,
  "bbox_codebook_size": 50,
  "bbox_codebook_dim": 128
}
```

```
config = {
  "max_stroke_len": 48,
  "udf_resolution": 64,
  "d_model": 256,
  "d_img": 128,
  "d_seq": 128,
  "embedding_dim": 256,
  "batch_size": 256,
  "learning_rate": 1e-4,
  "2-epochs": 1000,
  "n_head": 8,
  "num_layers": 6,
  "data_size_per_category": 50,
  "loss_weight": {
    "coord": 1.0,
    "pen": 0.1,
    "img": 0.1,
    "kl": 0.001
  },
}
```

Baseline (all codes are build from-scratch)

- VQ-VAE (Aaron 2018 et.al.)
- CNN, (Sketch former) (Leo 2020 et.al.)

Discrete Flow Matching

Baseline (all codes are build from-scratch)

- Backbone (GPT2(sketch) or U-net(image))
- corrector-scheduler, kappa-scheduler, probability path, CE-loss, source distribution, coupling is defined in specific combination

Mnist set-up

- Input: 32x32x3 flattened sequence (no pad)
- Arbitrary U-net (116M) -> outputs 32x32x3x257
- AdamW, lr=1e-4 (constant)
- Batch size: 64
- Cubic scheduler (t^3)

CFG in DFM

ICLR 2025

UNLOCKING GUIDANCE FOR DISCRETE STATE-SPACE DIFFUSION AND FLOW MODELS



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$$L_{cond}(x_i) = \log p(x_i|z, c)$$

Apply to logit space

$$L_{uncond}(x_i) = \log p(x_i|z, \emptyset)$$

$$L_{guided}(x_i) = \omega \cdot L_{cond}(x_i) + (1 - \omega) \cdot L_{uncond}(x_i)$$

$$p_{guided}(x_i) = \text{softmax}(L_{guided})_i$$

$$u_{guided}(x_i, z) = C_t \cdot (p_{guided}(x_i) - \delta_z(x_i))$$

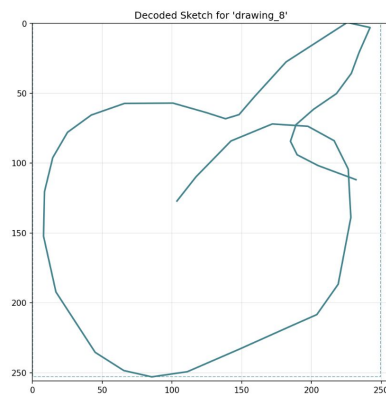
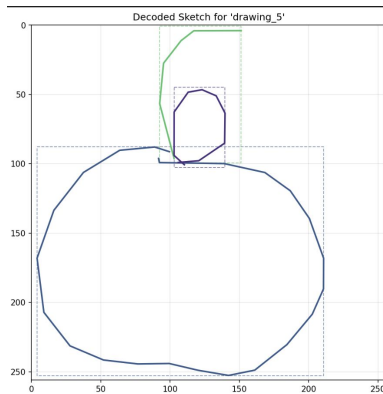
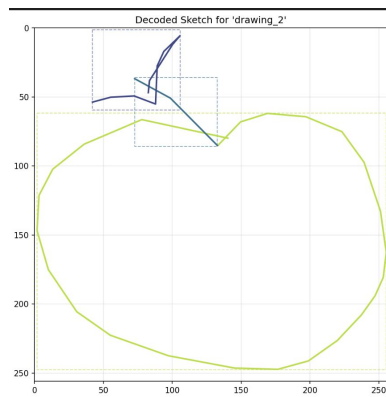
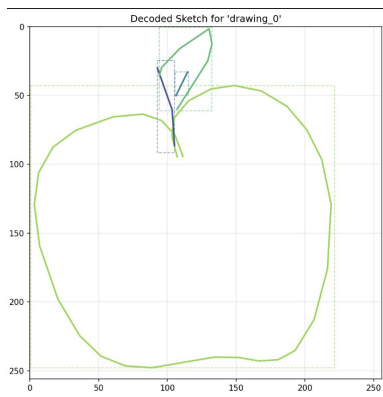
satisfy

$$p_{guided} = \frac{(p(x_i|z, c))^\omega \cdot (p(x_i|z, \emptyset))^{1-\omega}}{Z}$$

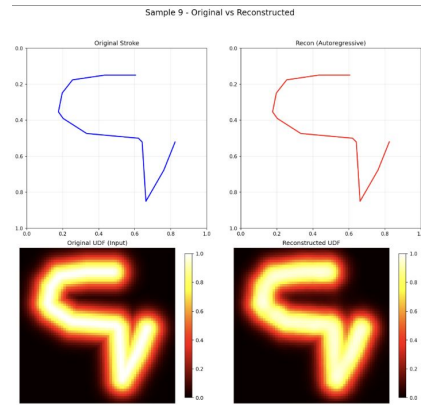
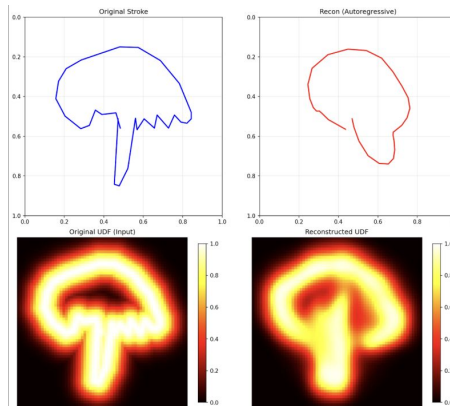
$$\sum_{x^i \in [d]} u_t^i(x^i, z) = 0, u_t^i(x^i, z) \geq 0, i \in [N] \text{ and } x^i \neq z^i$$

Flow matching sampling

Quantitative result



Cannot catch fine detail



Stroke reconstruction result

That's the reconstructed result (merged the bounding box, stroke)

Trained for small subset for quick-draw (2-category) <- overfitting

Future work

- (September) – Build MDM Architecture for ablation for DFM
 - Construct for the Creative sketch dataset (integration with label)
- (October) – Experiment with quick draw and Creative sketch dataset
 - Compare with other approaches
 - Build a Page to simulate each task